# CIAC — Chapter 02
# Large-signal & Small-signal Analysis

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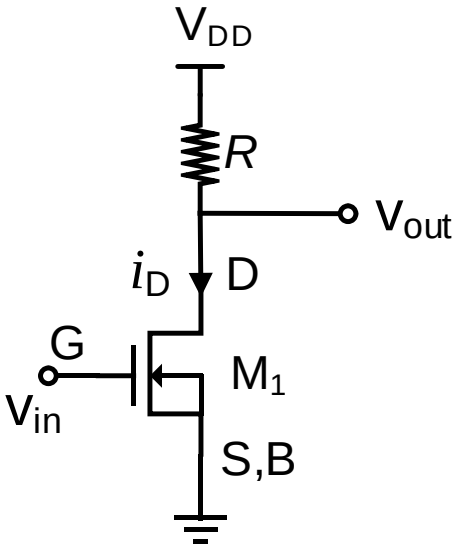
# Overview

- General Considerations
  - Operating / Bias Point
  - Principle of Superposition
  - From Large-signal to Small-signal Analysis
- MOS Transistor – Small-signal Model
  - Derivation of Small-signal Parameters
  - Small-signal Model with Capacitors

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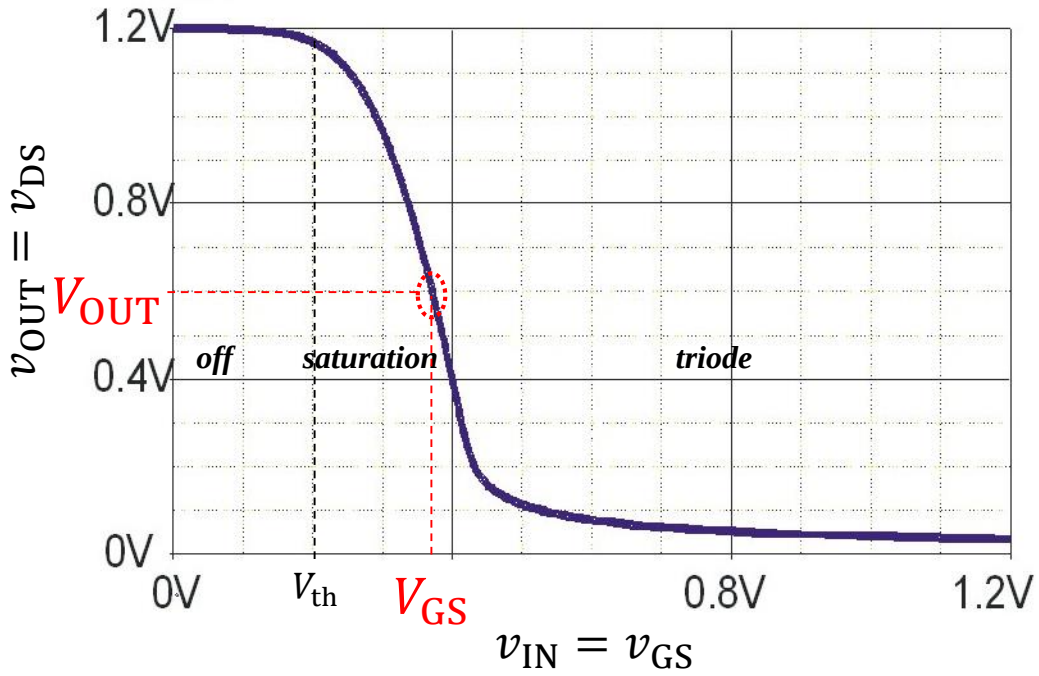
# Large-signal Behavior & Bias Point




inverter
with resistive load

$v_{IN} = v_{GS}$

$$v_{OUT} = v_{DS} = V_{DD} - R i_D$$




$0.12\ \mu\text{m CMOS}, W/L = 10/10$
$V_{DD} = 1.2\ \text{V}, R = 100\ \text{k}\Omega$

operating / bias point
$V_{OUT}(V_{IN} = V_{GS})$
for analog operation
(in saturation)

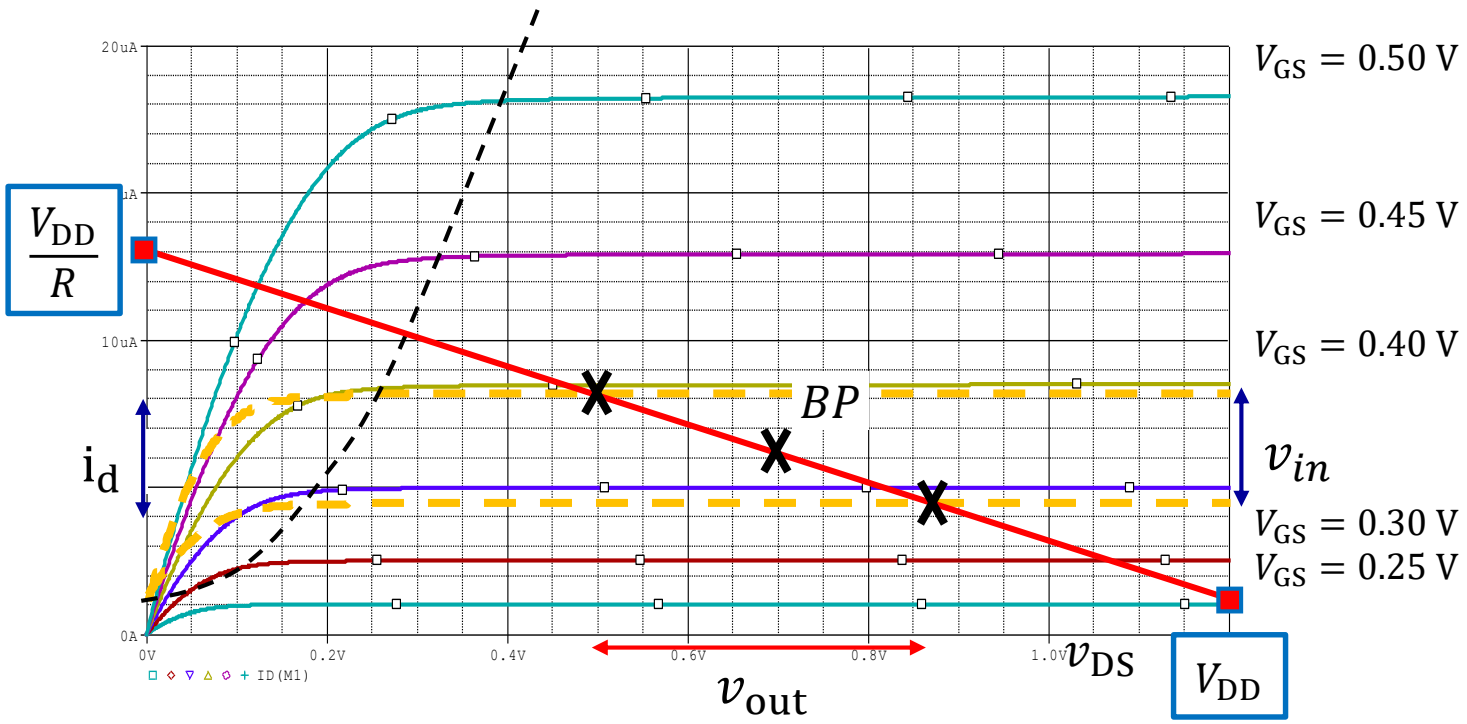
Inverter as a digital element: $v_{IN} = 0\ \text{V} | V_{DD} \rightarrow v_{OUT} = V_{DD} | 0\ \text{V}$

Inverter as an analog amplifier:
Define $I_D$ and thus $V_{OUT}$ by setting $V_{IN}$ for

- high amplification of small signal variations $v_{in}$ around $V_{IN}$
- high signal swing $v_{out}$ around $V_{OUT}$

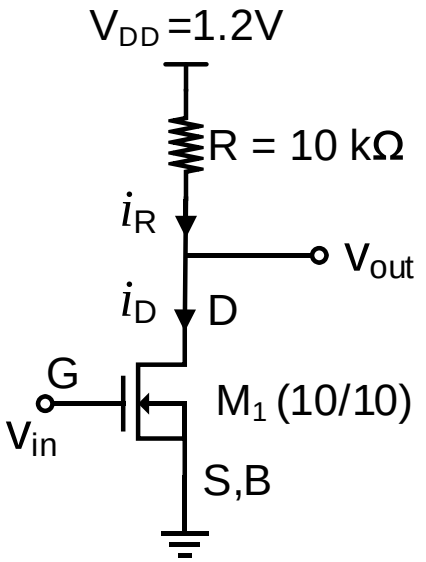
# Bias Point & Graphical Illustration of Amplification

$$i_D = \frac{V_{DD} - v_{DS}}{R}$$




$$i_D = \frac{\beta_n}{2}(v_{IN} - V_{th})^2$$
$$v_{OUT} = V_{DD} - Ri_D$$
$$i_D = i_R$$

$$i_d = G_m v_{in}$$
$$v_{out} = -Ri_d$$
$$= -G_m R v_{in}$$




$$v_{in} = v_{gs} \qquad v_{in} \longrightarrow i_d$$
$$v_{out} = v_{ds} \qquad i_d \longrightarrow v_{out}$$

$$A = \frac{v_{out}}{v_{in}}$$

transconductance

$$G_m = \frac{i_{out}}{v_{in}} = \frac{i_d}{v_{gs}}$$

$$A = \frac{v_{out}}{v_{in}} = -G_m R$$

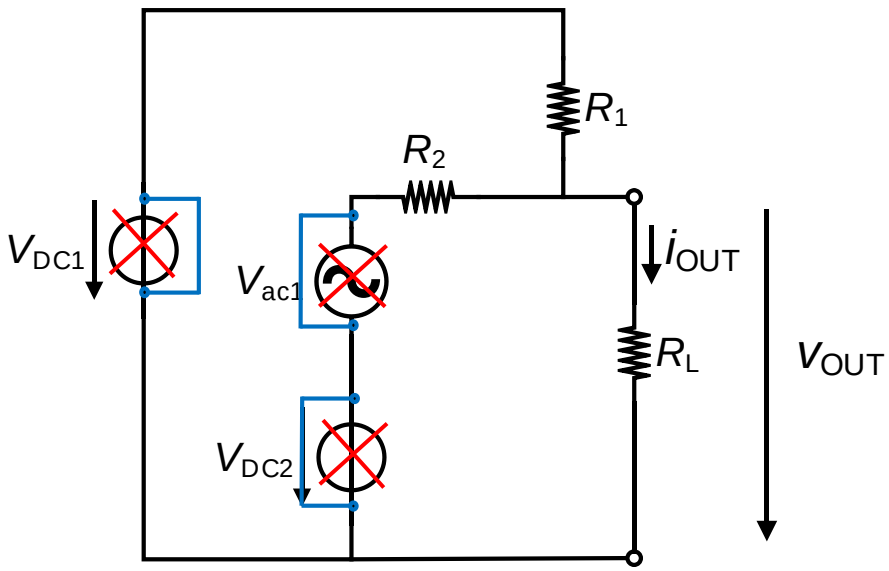
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# Superposition in Linear Networks

In a linear network, the principle of superposition is applicable!

example:




$$v_{OUT} = f_1(V_{DC1})|_{V_{DC2} = v_{ac1} = 0\,V} = \frac{R_L||R_2}{R_1 + R_L||R_2} V_{DC1}$$

$$+ f_2(V_{DC2})|_{V_{DC1} = v_{ac1} = 0\,V} + \frac{R_L||R_1}{R_2 + R_L||R_1} V_{DC2}$$

$$+ f_3(v_{ac1})|_{V_{DC1} = V_{DC2} = 0\,V} + \frac{R_L||R_1}{R_2 + R_L||R_1} v_{ac1}$$

$$i_{OUT} = \frac{v_{OUT}}{R_L} = i_{OUT1}(V_{DC1})|_{V_{DC2} = v_{ac1} = 0\,V}$$

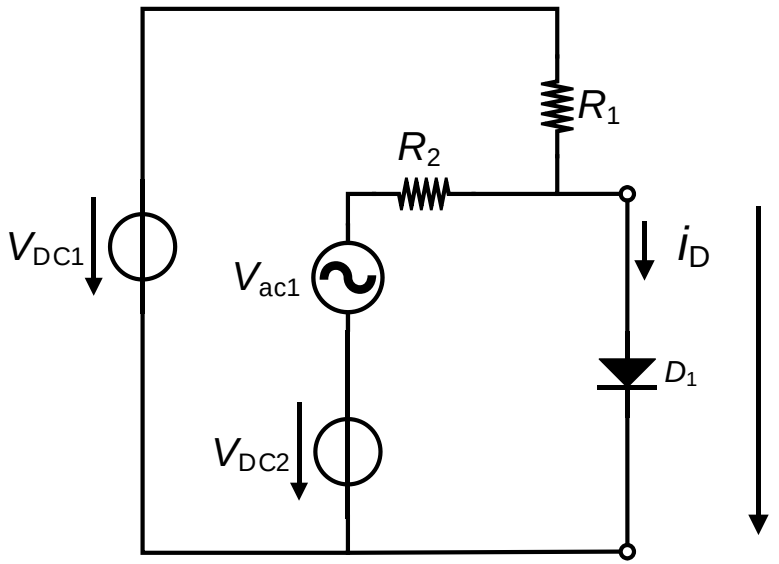
$$+ i_{OUT2}(V_{DC2})|_{V_{DC1} = v_{ac1} = 0\,V}$$

$$+ i_{OUT3}(v_{ac1})|_{V_{DC1} = V_{DC2} = 0\,V}$$

# Non-linear Networks

In a non-linear network, the principle of superposition is <u>not</u> applicable!

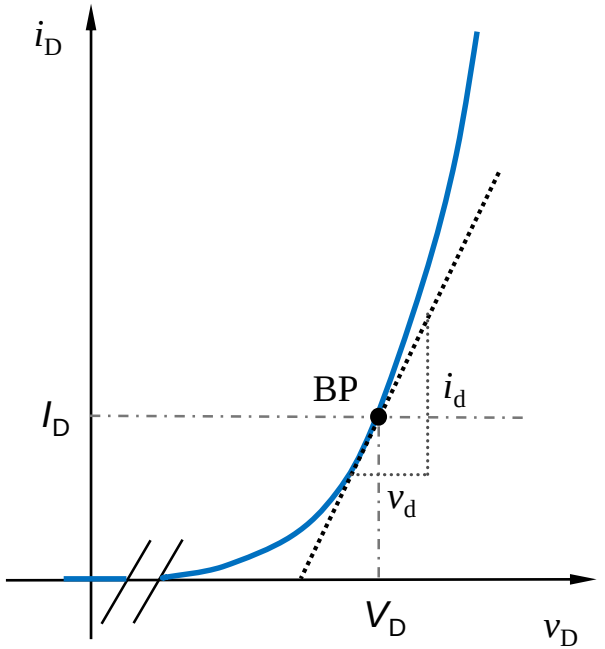
example:




$i_D(v_D)$-characteristic of a diode:

$$i_D = I_S\left(e^{\frac{v_D}{V_T}} - 1\right)$$

$$\neq \; i_{D1}(V_{DC1})\big|_{V_{DC2}=v_{ac1}=0\,V} + i_{D2}(V_{DC2})\big|_{V_{DC1}=v_{ac1}=0\,V} + i_{D3}(v_{ac1})\big|_{V_{DC1}=V_{DC2}=0\,V}$$




Here, a linear approximation is performed by means of a

large-signal & small-signal analysis

according to a first-order Taylor series approximation of the non-linear circuit elements!

# Overview

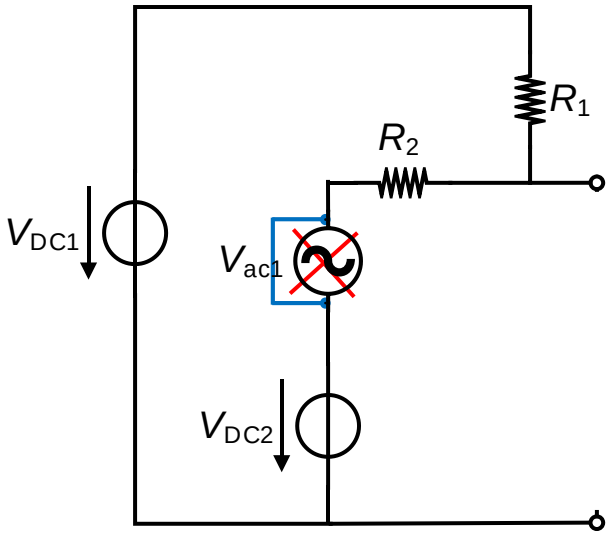
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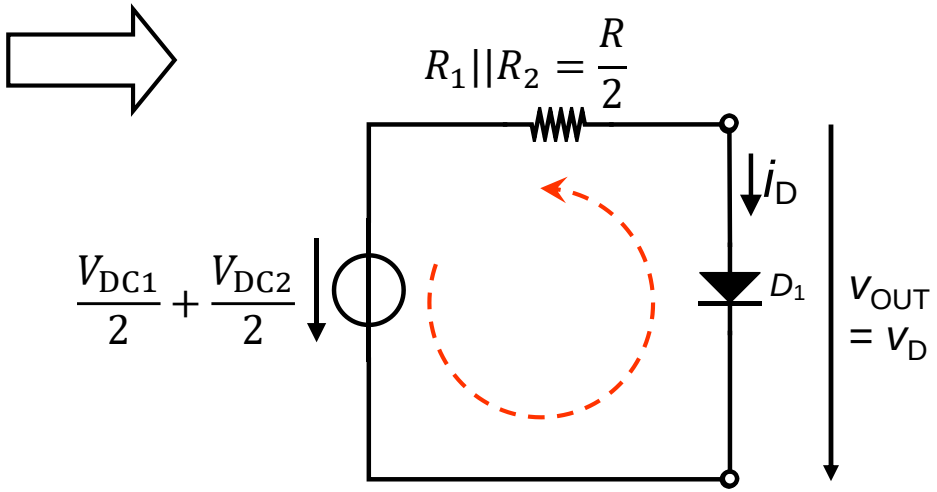
# Step 1 – Large-signal Analysis

Calculation of the bias point

Consider all DC current and voltage sources while neglecting all AC sources!

example: $R_1 = R_2 = R$




Thévenin equivalent

Kirchhoff's voltage loop

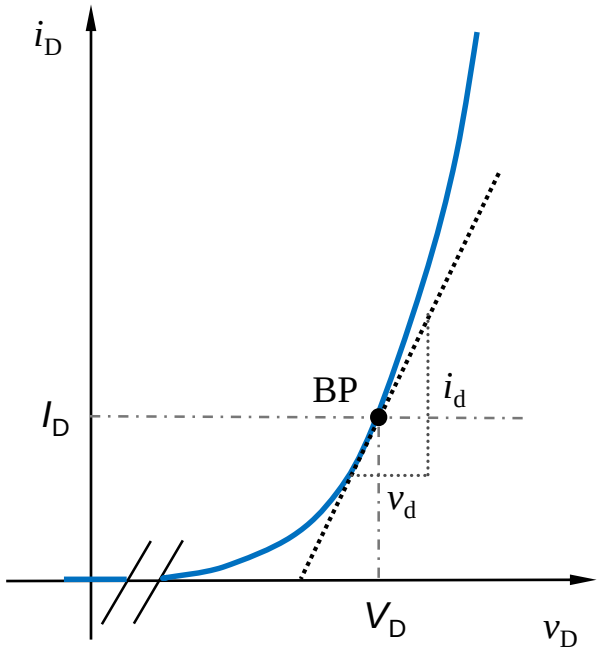
$$\frac{V_{DC1}}{2} + \frac{V_{DC2}}{2} - v_D - \frac{R}{2} i_D = 0$$

$$\frac{V_{DC1}}{2} + \frac{V_{DC2}}{2} = \frac{R}{2}\left(I_S e^{\frac{q v_D}{kT}} - 1\right) + v_D$$

Solving for $v_D$ results in the bias point information

$$\rightarrow \boldsymbol{V_D, I_D}$$

# Step 2 – Small-signal Analysis




Based on the bias point, a first-order **Taylor series approximation** for the diode's non-linear $i_D(v_D)$-characteristic can be performed.

$i_D(v_D)$-characteristic: $i_D = I_S(e^{\frac{qv_D}{kT}} - 1) \approx I_S e^{\frac{qv_D}{kT}}$ for $\frac{qv_D}{kT} \gg 0$

first-order **Taylor series approximation**

$$i_D \approx i_D(V_D) + \left.\frac{\partial i_D}{\partial v_D}\right|_{V_D} (v_D - V_D) = I_D + g_d v_d$$

bias point (large-signal analysis), already calculated

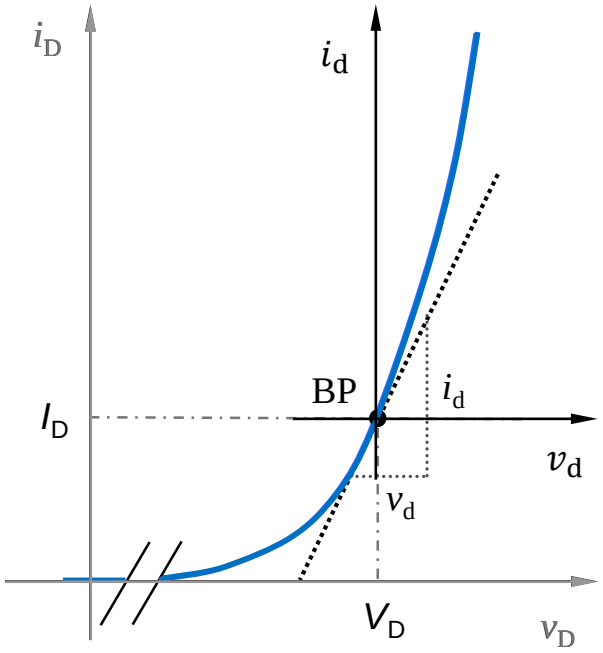
slope of the tangent to the diode's $i_D(v_D)$-characteristic at the bias point (small-signal parameter)

small-signal part $v_d$ of $v_D$ (small-signal analysis, still to be performed)

# Small-signal Analysis – Reference Point

Step 2: **Small-signal analysis**

As soon as the bias point and thus the slope of the straight line tangent to the non-linear I/V characteristic is known, the bias point is of no further interest for the small signal analysis!




For this reason, the circuit is “moved” into the bias point whereby a new reference point is defined for the small signal analysis. We say the point is at “virtual ground”.

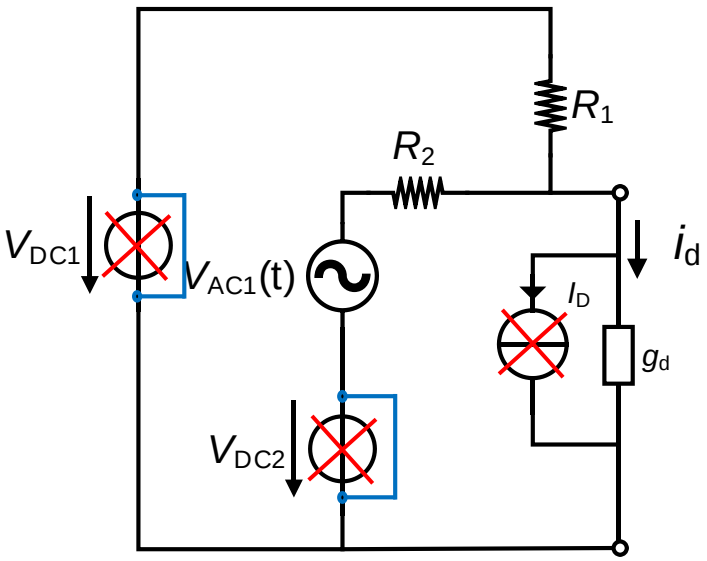
Speaking in circuit terms, this is equivalent to removing all DC sources.

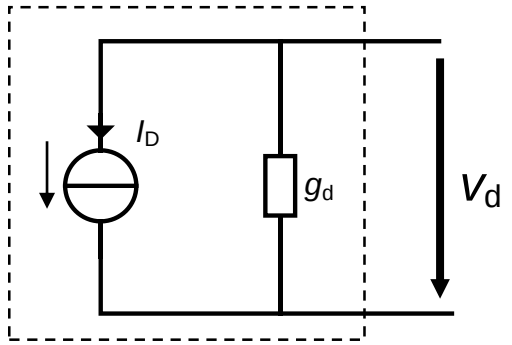
(DC voltage sources are to be shorted, DC current sources are to be disconnected.)

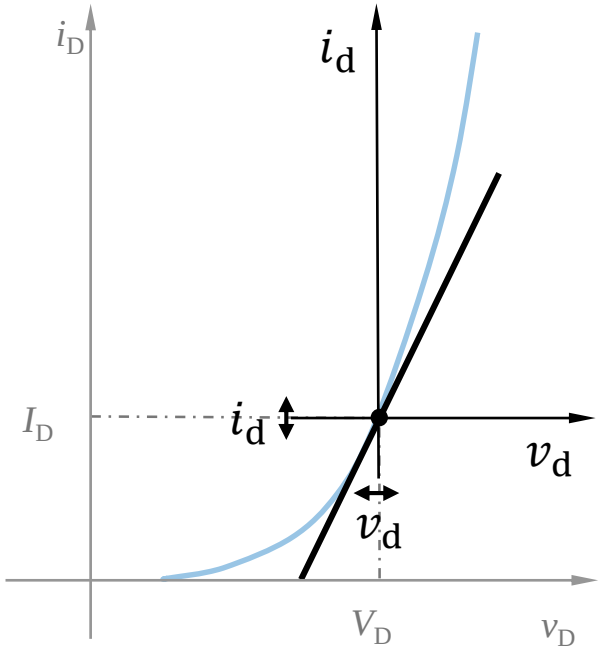
# Small-signal Analysis – Equivalent Model

constant conductance $g_\mathrm{d}$

$$i_\mathrm{D}(t) = \underbrace{i_\mathrm{D}(V_\mathrm{D})}_{I_\mathrm{D}} + \overbrace{\left.\frac{\partial i_\mathrm{D}}{\partial v_\mathrm{D}}\right|_{V_\mathrm{D}}}\ \underbrace{(v_\mathrm{D}(t) - V_\mathrm{D})}_{\text{small-signal part } v_\mathrm{d}(t)} = I_\mathrm{D} + g_\mathrm{d} v_\mathrm{d}(t)$$



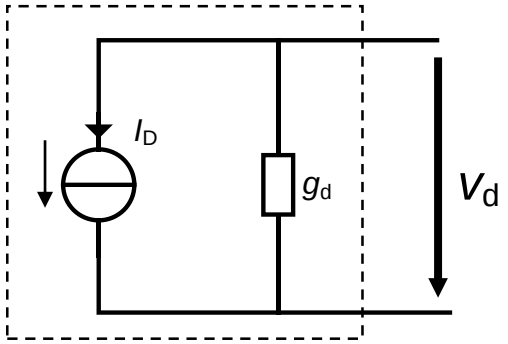



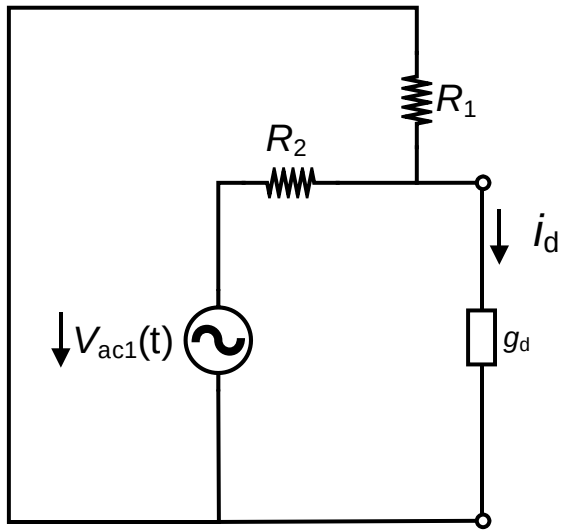



# Small-signal Analysis – Equivalent Model

constant conductance $g_\mathrm{d}$

$$i_\mathrm{D}(t) = \underbrace{i_\mathrm{D}(V_\mathrm{D})}_{I_\mathrm{D}} + \underbrace{\frac{\partial i_\mathrm{D}}{\partial v_\mathrm{D}}\Big|_{V_\mathrm{D}}}_{} \underbrace{(v_\mathrm{D}(t) - V_\mathrm{D})}_{\text{small-signal part } v_\mathrm{d}(t)} = I_\mathrm{D} + g_\mathrm{d} v_\mathrm{d}(t)$$

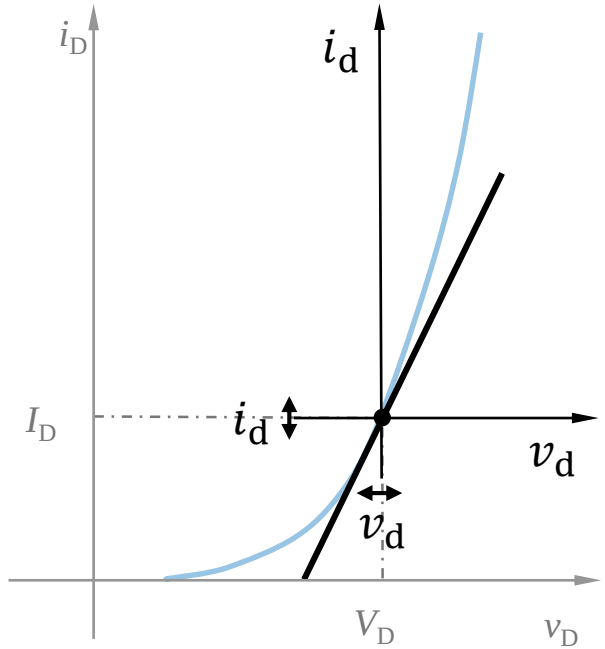



equivalent linear small-signal model

$$v_\mathrm{d}(t) = \frac{R_1 \parallel g_\mathrm{d}}{R_2 + R_1 \parallel g_\mathrm{d}} v_\mathrm{ac1}(t)$$

$v_\mathrm{ac1}$(t) is supposed to be a *small* AC signal!



# Summary

- The principle of superposition **is not** applicable to non-linear circuits!
- In order to perform a small-signal analysis, a large-signal analysis (operating / bias point) must be performed first, yielding the small-signal parameters!
- The principle of superposition **is** applicable to a small-signal analysis since the non-linear network has been approximated by a linear one!
- DC sources are of no interest for the small-signal analysis!
  For this reason
  - short any DC voltage source $\frac{\partial V_{\text{constant}}}{\partial t} = 0 \longrightarrow R_{\text{equivalent}} = 0\ \Omega$
  - open any branch with a DC current source $\frac{\partial I_{\text{constant}}}{\partial t} = 0 \longrightarrow R_{\text{equivalent}} = \infty\ \Omega$

The circuit's reference voltage (ground) is thus moved onto the bias point!

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# Large-signal Equations

**subthreshold region** $v_{GS} < V_{th}$

$i_D \approx 0\ \mathrm{A}$

valid for channels with *W*, *L* > 1µm

**linear region** $v_{GS} > V_{th}$ && $v_{DS} < v_{GS} - V_{th}$

$$i_D = \beta \left[ (v_{GS} - V_{th}) v_{DS} - \frac{1}{2} v_{DS}^2 \right] (1 + \lambda v_{DS})$$

**saturation region** $v_{GS} > V_{th}$ && $v_{DS} \geq v_{GS} - V_{th}$

$$i_D = \frac{\beta}{2} (v_{GS} - V_{th})^2 (1 + \lambda v_{DS})$$

with $\beta = \beta_0 \frac{W}{L} = \mu_n C'_{ox},\ C'_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$

$$V_{th} = V_{th,0} + \gamma \left( \sqrt{2\Phi_F + v_{SB}} - \sqrt{2\Phi_F} \right)$$

# Small-signal Parameters

linear region

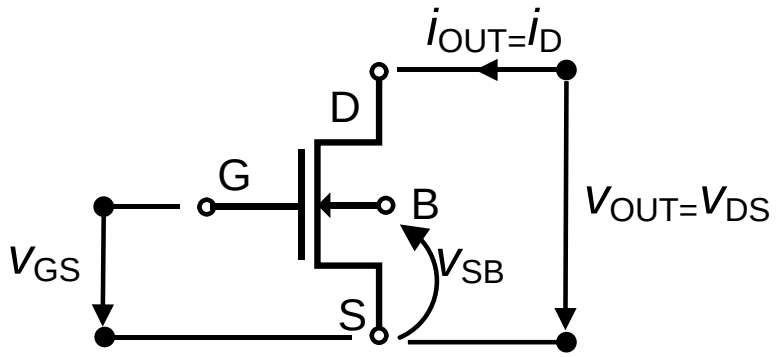
non-linear functions

saturation region

$$i_D = \beta\left[(v_{GS} - V_{th})v_{DS} - \frac{1}{2}v_{DS}^2\right](1 + \lambda v_{DS})$$

$$i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2\,(1 + \lambda v_{DS})$$

$$V_{th} = V_{th,0} + \gamma\left(\sqrt{2\Phi_F + v_{SB}} - \sqrt{2\Phi_F}\right)$$




$$i_D = f(v_{GS}, v_{DS}, v_{SB})$$

$$\frac{\partial i_D}{\partial v_{GS}} = g_m$$

transconductance due to $v_{GS}$

$$\frac{\partial i_D}{\partial v_{DS}} = g_{ds}$$

output conductance due to $v_{DS}$

$$\frac{\partial i_D}{\partial v_{BS}} = \frac{\partial i_D}{\partial V_{th}}\frac{\partial V_{th}}{\partial v_{BS}} = g_{mb}$$

transconductance due to $v_{BS}$

# Taylor Approximation

first-order Taylor approximation: $i_\mathrm{D} = I_\mathrm{D} + i_\mathrm{d}$ $\quad v_\mathrm{GS} = V_\mathrm{GS} + v_\mathrm{gs}$

$v_\mathrm{DS} = V_\mathrm{DS} + v_\mathrm{ds}$ $\quad v_\mathrm{SB} = V_\mathrm{SB} + v_\mathrm{sb}$

large signals (bias point): $I_\mathrm{D}, V_\mathrm{GS}, V_\mathrm{DS}, V_\mathrm{BS}$

small signals: $i_\mathrm{d}, v_\mathrm{gs}, v_\mathrm{ds}, v_\mathrm{sb}$

$$i_\mathrm{D} = I_\mathrm{D} + i_\mathrm{d}$$

large signal (bias point) $\qquad$ small signal

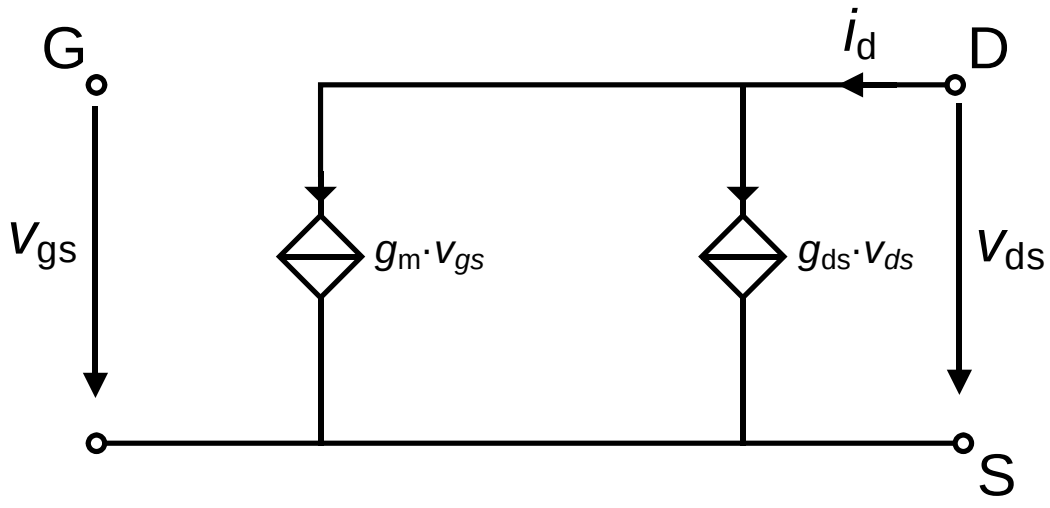
$$I_\mathrm{D} = i_\mathrm{D}\Big|_{BP}$$

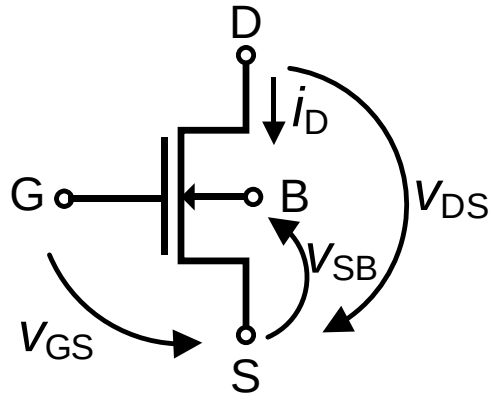
$$i_\mathrm{d} = \frac{\partial i_\mathrm{D}}{\partial v_\mathrm{GS}}\Big|_{BP}(v_\mathrm{GS} - V_\mathrm{GS}) + \frac{\partial i_\mathrm{D}}{\partial v_\mathrm{DS}}\Big|_{BP}(v_\mathrm{DS} - V_\mathrm{DS}) + \frac{\partial i_\mathrm{D}}{\partial v_\mathrm{BS}}\Big|_{BP}(v_\mathrm{BS} - V_\mathrm{BS})$$

$$= g_\mathrm{m} v_\mathrm{gs} + g_\mathrm{ds} v_\mathrm{ds} + g_\mathrm{mb} v_\mathrm{bs}$$

linear circuit

# nMOS Transistor – Small-signal Model




$$g_m = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{BP}$$

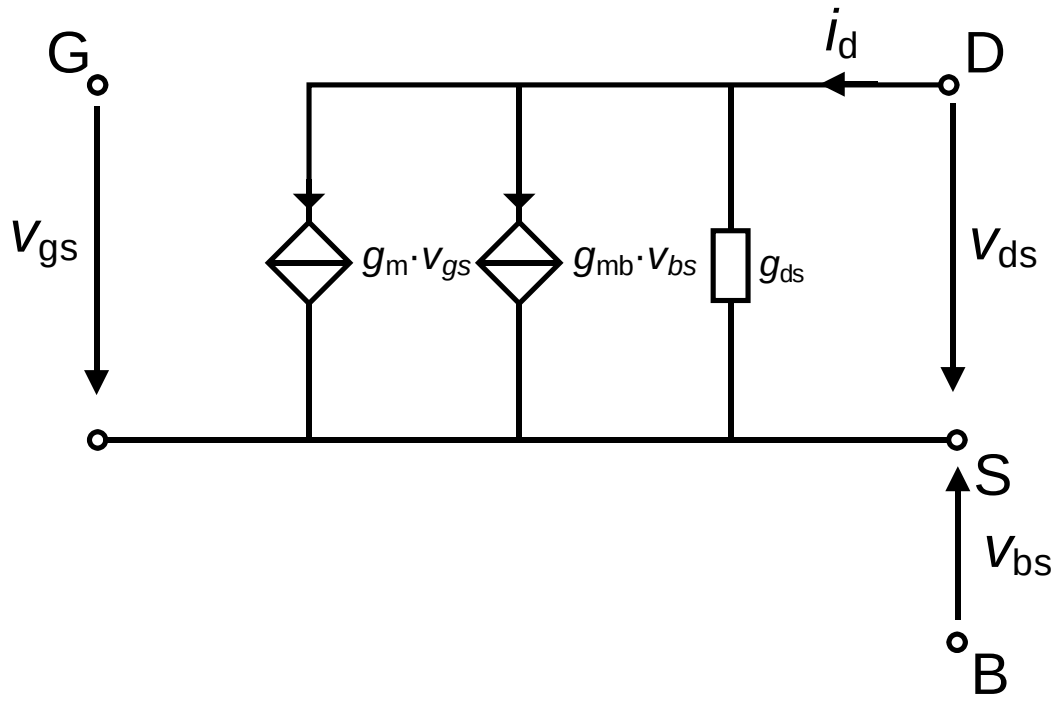
transconductance due to $v_{gs}$

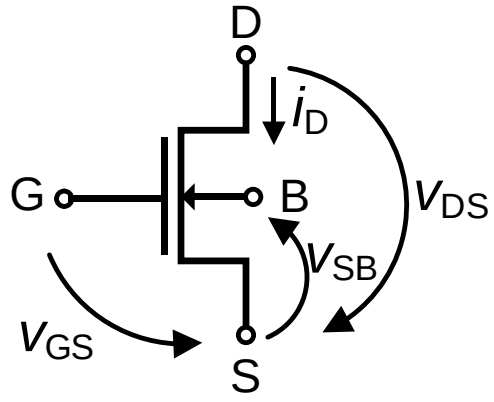
voltage-controlled current source

$$g_{ds} = \left.\frac{\partial i_D}{\partial v_{DS}}\right|_{BP}$$

output conductance due to $v_{ds}$

# nMOS Transistor – Small-signal Model




$$g_m = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{BP}$$

transconductance due to $v_{gs}$

voltage-controlled current source

$$g_{ds} = \left.\frac{\partial i_D}{\partial v_{DS}}\right|_{BP}$$

output conductance due to $v_{ds}$

$$g_{mb} = \left.\frac{\partial i_D}{\partial v_{BS}}\right|_{BP} = \left.\frac{\partial i_D}{\partial V_{th}}\frac{\partial V_{th}}{\partial v_{BS}}\right|_{BP}$$

transconductance due to $v_{bs}$

# Small-signal Parameters – Saturation

large-signal equations

$$i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2 (1 + \lambda v_{DS}) \qquad V_{th} = V_{th,0} + \gamma\left(\sqrt{2\Phi_F + v_{SB}} - \sqrt{2\Phi_F}\right)$$

transconductance due to $v_{GS}$

$$g_m = \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{BP} \approx \beta(V_{GS} - V_{th}) = \sqrt{2\beta I_D} = \sqrt{2\beta_0 \frac{W}{L} I_D}$$

large-signal parameter

small-signal parameter

output conductance due to $v_{DS}$

$$g_{ds} = \left.\frac{\partial i_D}{\partial v_{DS}}\right|_{BP} = \lambda\frac{\beta}{2}(V_{GS} - V_{th})^2 \quad \frac{1 + \lambda V_{DS}}{1 + \lambda V_{DS}} = \frac{\lambda I_D}{1 + \lambda V_{DS}} \approx \lambda I_D$$

output resistance

$$r_{ds} \approx \frac{1}{\lambda I_D}$$

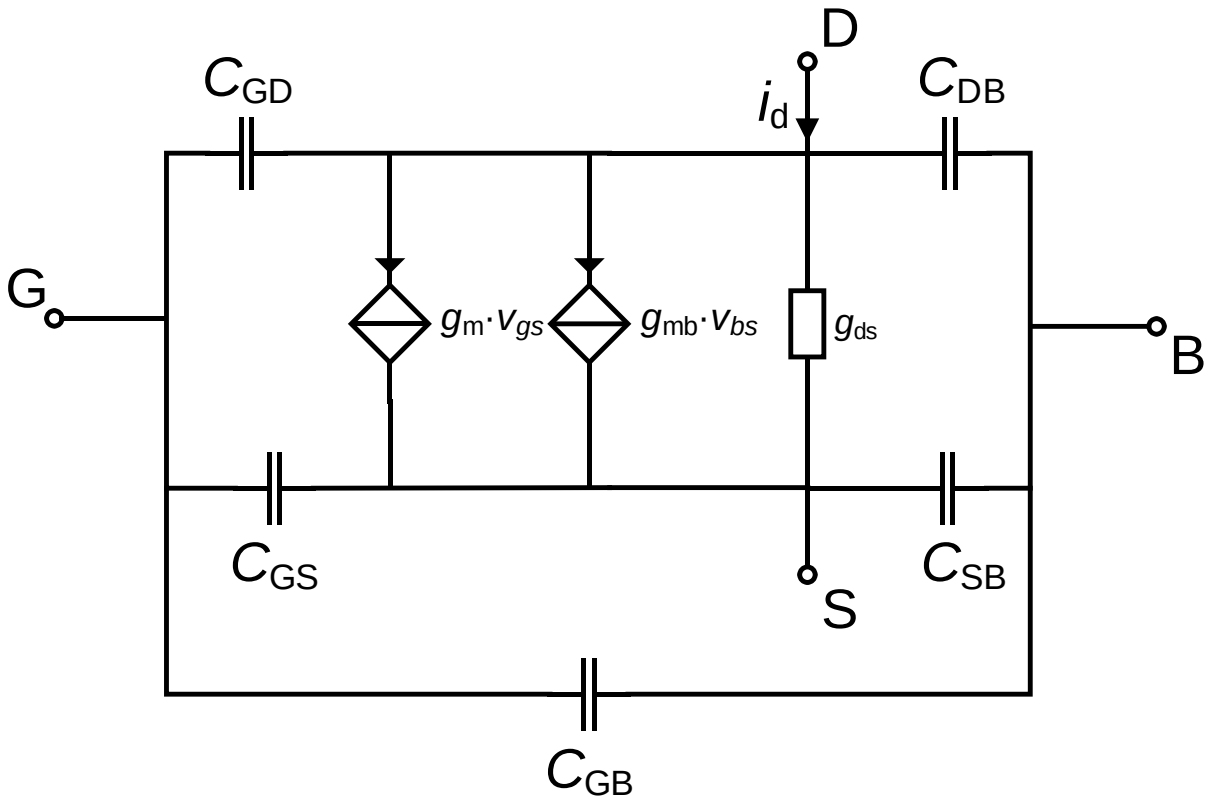
transconductance due to $v_{BS}$

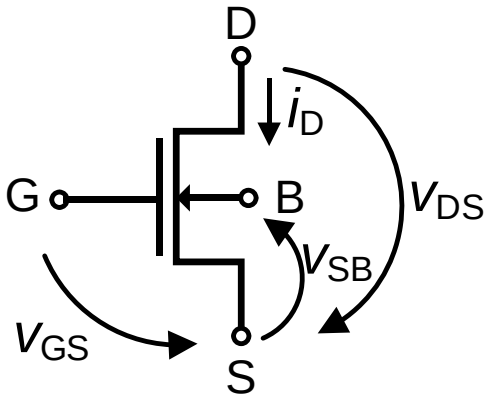
$$g_{mb} = \left.\frac{\partial i_D}{\partial v_{BS}}\right|_{BP} = \left.\frac{\partial i_D}{\partial V_{th}}\frac{\partial V_{th}}{\partial v_{BS}}\right|_{BP} = -g_m \left.\frac{\partial V_{th}}{\partial v_{BS}}\right|_{BP} = g_m \left.\frac{\partial V_{th}}{\partial v_{SB}}\right|_{BP} = g_m \frac{\gamma}{2\sqrt{2\Phi_F + V_{SB}}} = \eta g_m$$

# Overview

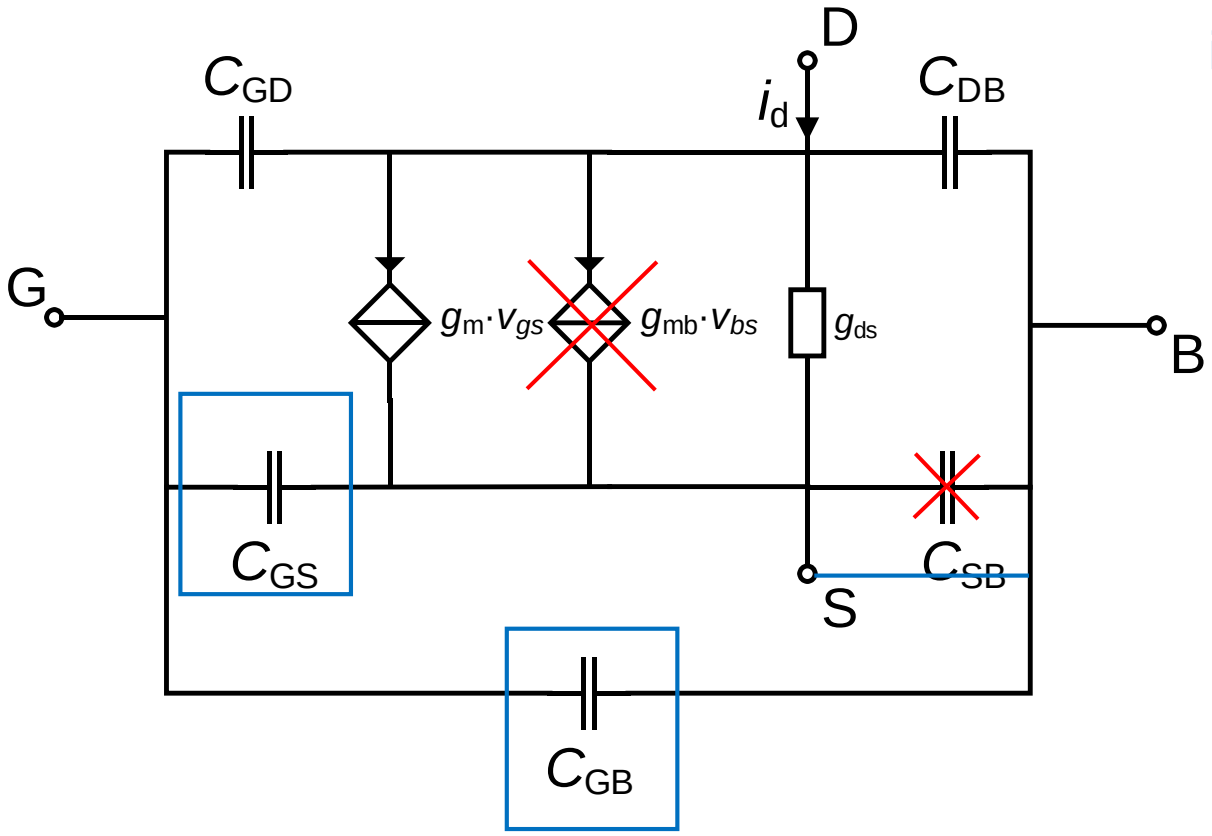
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# Small-signal Model with Capacitors

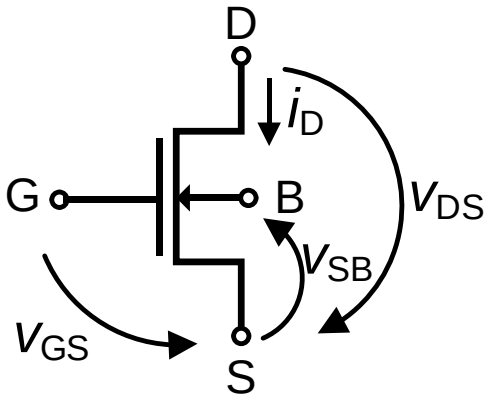


# Small-signal Model with $v_{SB} = 0$ V



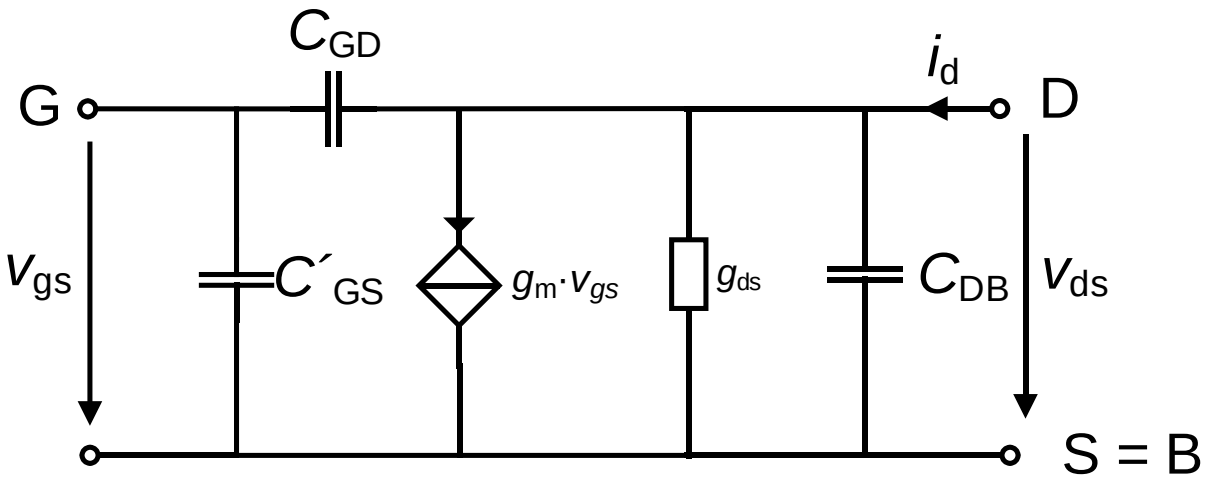

if $v_{SB} = 0$ V

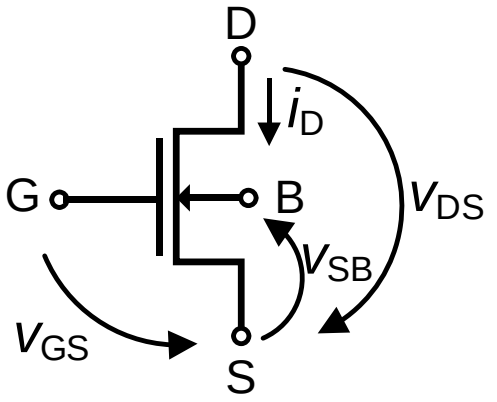



$$C'_{GS} = C_{GS} + C_{GB}$$

# Simplified Small-signal Model with $v_{SB} = 0\ \mathrm{V}$

special case: $v_{SB} = 0\ \mathrm{V}$




$$C'_{GS} = C_{GS}(+C_{GB})$$

# Small-signal Model – Typical Parameter Values

saturation ($V_{GS}$ = 350 mV, $V_{th} \approx$ 200 mV, $V_{DS}$ = 400 mV, $V_{SB}$ = 0 V)

| | $W/L$ = 10/10 | $W/L$ = 1/10 | $W/L$ = 10/1 | $W/L$ = 1/1 | $W/L$ = 1/0.5 |
|---|---|---|---|---|---|
| $g_m$ | 57.2 µA/V | 5.3 µA/V | 511 µA/V | 47.4 µA/V | 80 µA/V |
| $g_{ds}$ | 0.16 µS | 0.014 µS | 6.4 µS | 0.57 µS | 1.37 µS |
| $C_{GS}$ | 840 fF | 84.2 fF | 87.9 fF | 8.8 fF | 4.4 fF |
| $C_{GD}$ | 4.6 fF | 0.46 fF | 3 fF | 0.3 fF | 0.3 fF |
| $C_{DB}$ | same order of magnitude as $C_{GD}$ (min. 10x smaller than $C_{GS}$) | | | | |

# Small-signal Model – Summary

saturation region

$$i_{\mathrm{D}} = \frac{\beta_0}{2}\frac{W}{L}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2$$

$$g_{\mathrm{ds}} = \left.\frac{\partial i_{\mathrm{D}}}{\partial v_{\mathrm{DS}}}\right|_{BP} \approx \lambda I_{\mathrm{D}}$$

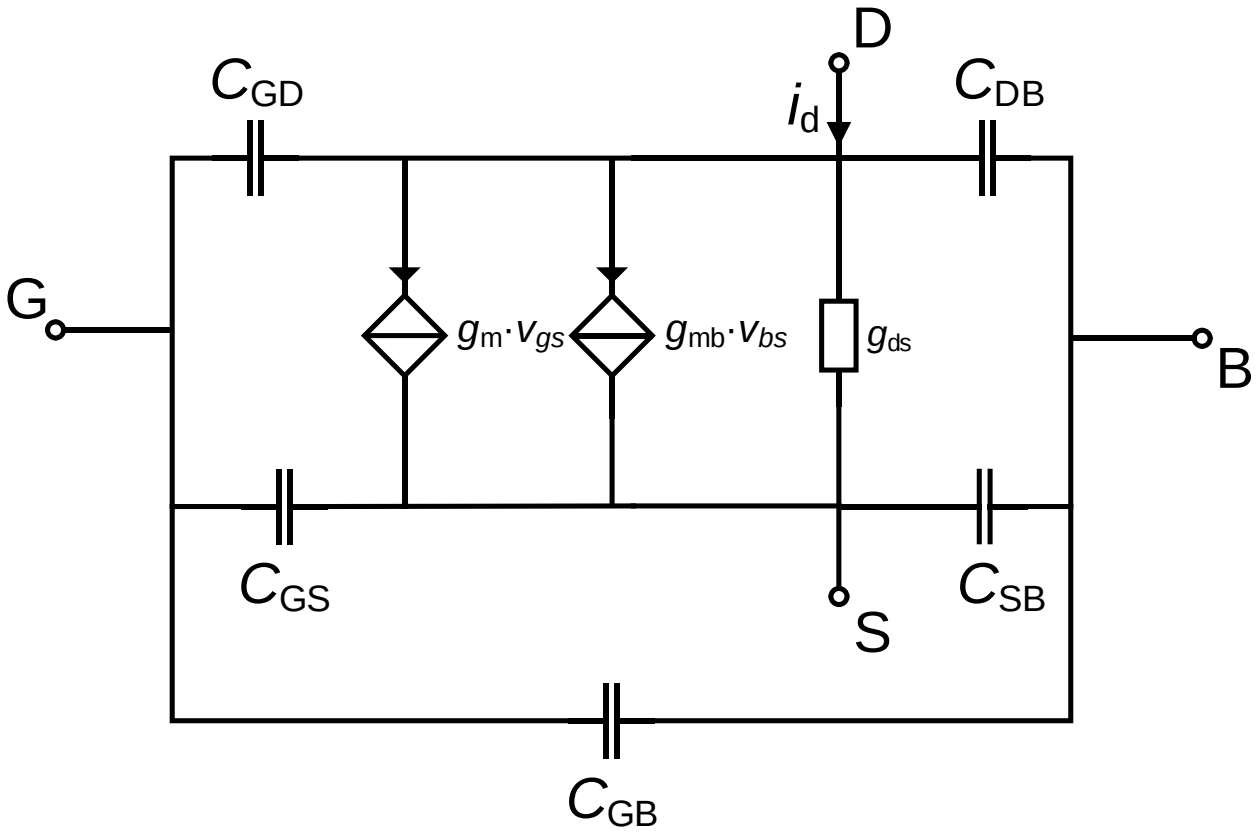
$$g_{\mathrm{m}} = \left.\frac{\partial i_{\mathrm{D}}}{\partial v_{\mathrm{GS}}}\right|_{BP} = \beta_0 \frac{W}{L}(V_{\mathrm{GS}} - V_{\mathrm{th}})$$

linear region

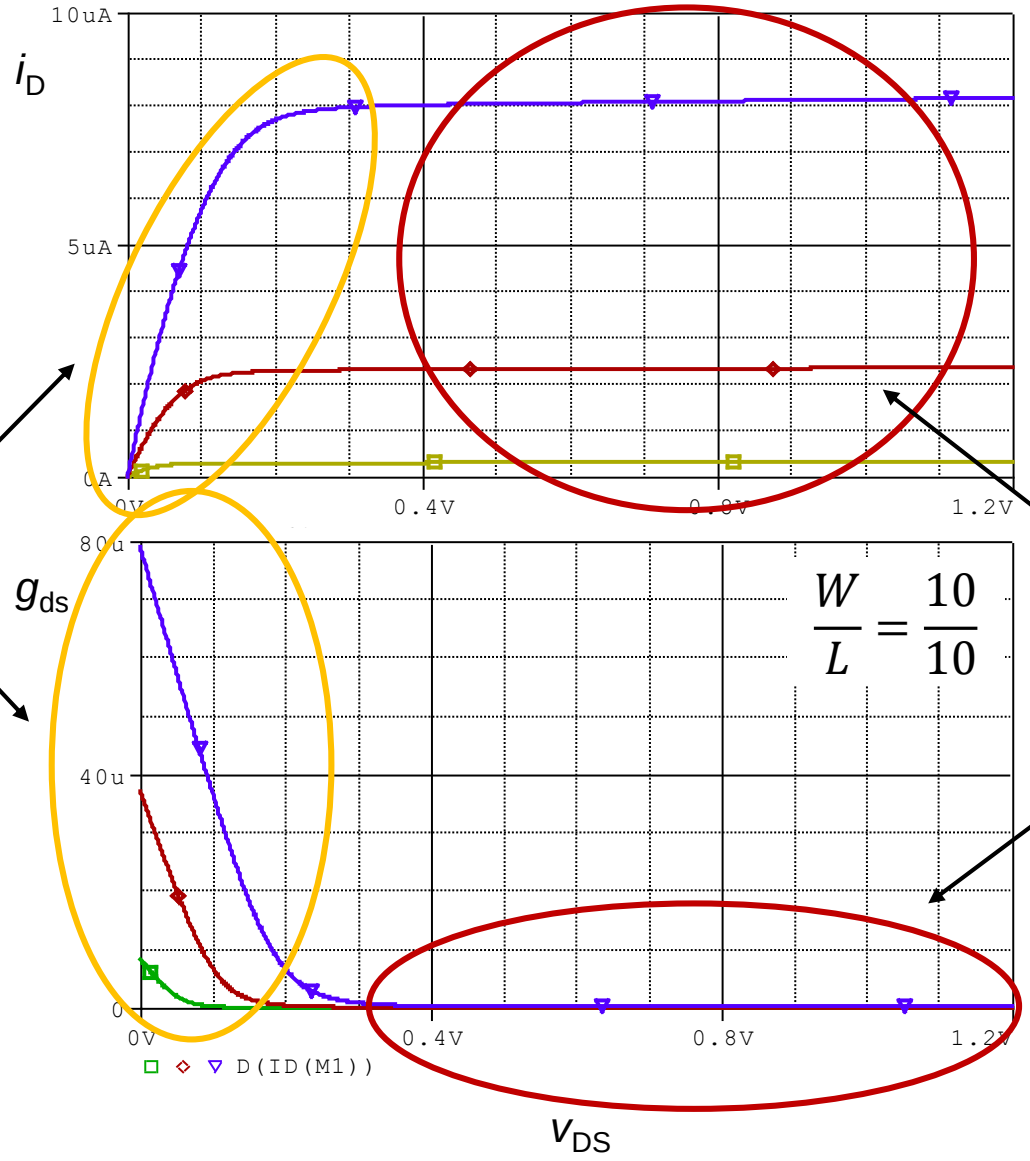
$$i_{\mathrm{D}} = \beta_0 \frac{W}{L}\left(v_{\mathrm{GS}} - V_{\mathrm{th}} - \frac{v_{\mathrm{DS}}}{2}\right) v_{\mathrm{Ds}}$$

$$g_{\mathrm{ds}} = \left.\frac{\partial i_{\mathrm{D}}}{\partial v_{\mathrm{DS}}}\right|_{BP} = \beta_0 \frac{W}{L}\left(v_{\mathrm{GS}} - V_{\mathrm{th}} - \frac{v_{\mathrm{DS}}}{2}\right)$$

$$g_{\mathrm{m}} = \left.\frac{\partial i_{\mathrm{D}}}{\partial v_{\mathrm{GS}}}\right|_{BP} = \beta_0 \frac{W}{L} V_{\mathrm{DS}}$$



# Small-signal Conductance $g_{ds}$




$V_{GS}$ = 400 mV

$V_{GS}$ = 300 mV

$V_{GS}$ = 200 mV

## linear region

$i_D \propto v_{DS}^2$

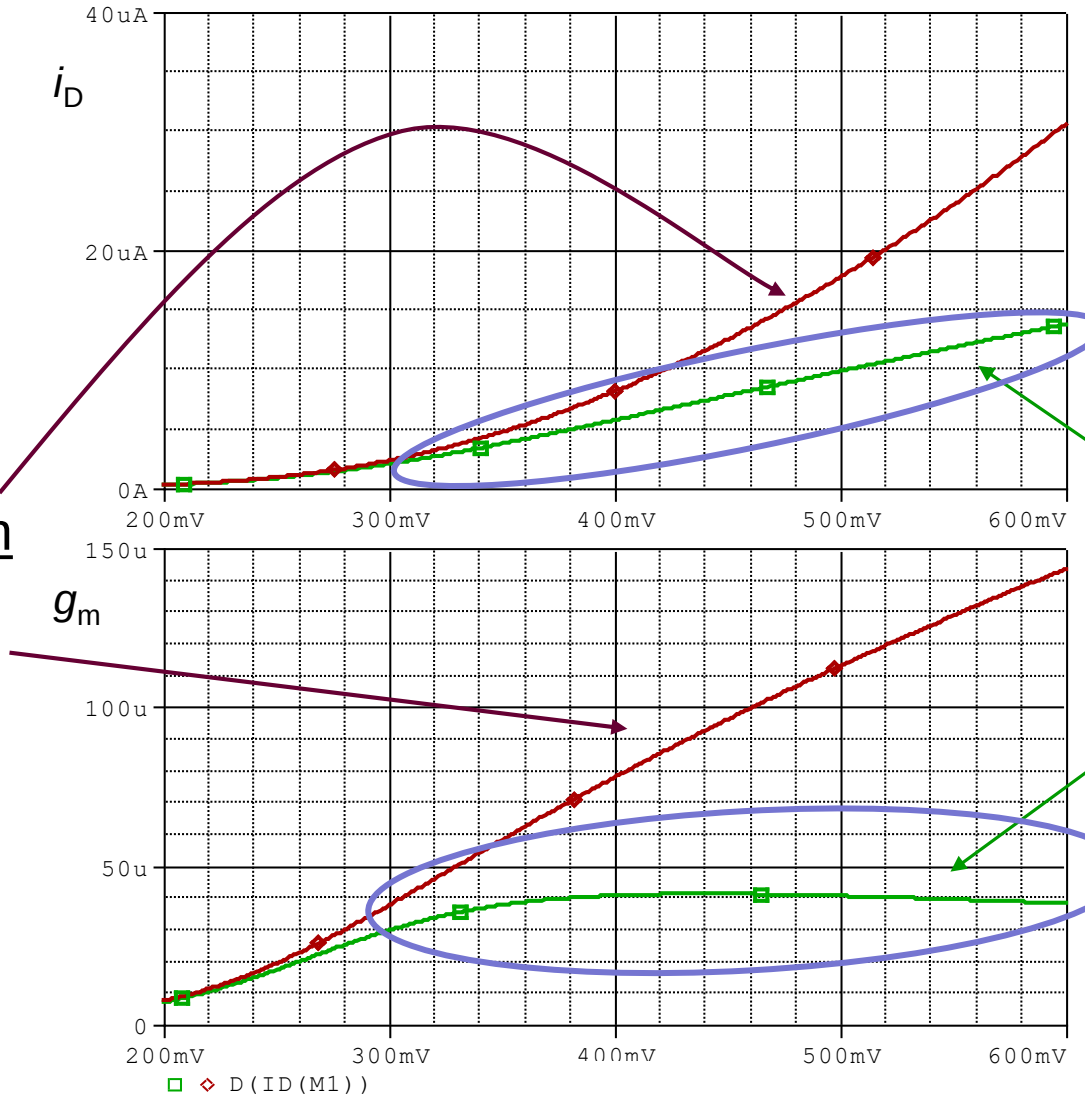
$g_{ds} \propto v_{DS}$

## saturation region

$i_D$ almost constant $\neq f(v_{DS})$

$g_{ds} \approx 0\ \mathrm{S}\ (r_{ds} \rightarrow \infty\ \Omega)$

# Small-signal Transconductance $g_m$




$V_{DS}$ = 1.2 V

$V_{DS}$ = 100 mV

saturation region

$i_D \propto v_{GS}^2$

$g_m \propto v_{DS}$

linear region

$i_D \propto v_{GS}$

$g_m$ constant $\neq f(v_{GS})$

# Summary

- Small-signal parameters are used to approximate the MOS behavior for “small” signal variations.
- Small-signal parameters depend on the operating region of the MOS transistor.
- Small-signal parameters depend on the operating point & thus on large-signal voltages and currents.

IMPORTANT to understand:

Even for small signals, the transistor still behaves according to the large-signal equations ($i_D \propto v_{GS}^2$ in saturation).

→ However, we assume a LINEAR behavior AT THE OPERATING POINT for very, very small signals (Taylor approximation).

# Literature

- Razavi, chapter 2.4.3, pp. 33-36
- Sedra / Smith, chapter 4.6, pp. 287-298
- Allen / Holberg, chapter 3.3, pp. 87-92